

# Linear Algebra: linear functionals

Chong Jongwon

June 30, 2026

## Contents

|                                                                |    |
|----------------------------------------------------------------|----|
| 1 Dual Space                                                   | 2  |
| 2 Double Dual                                                  | 4  |
| 3 Transpose                                                    | 5  |
| 4 Hyperspace                                                   | 6  |
| 5 Annihilators                                                 | 7  |
| 6 Properties of Annihilators                                   | 8  |
| 7 Transpose and Annihilators                                   | 9  |
| 8 System of Linear Equations in the View of Linear functionals | 10 |
| 9 Inner Product Space and Linear functional                    | 11 |

# 1 Dual Space

Let  $V$  be a vector space over a field  $F$ .

**Definition 1.1.** A linear map  $f : V \rightarrow F$  is called a *linear functional* on  $V$ .

**Example.** Let  $F[x]$  be the set of polynomials over  $F$ . For each  $\alpha \in F$ , define

$$L_\alpha : F[x] \rightarrow F : p(x) \mapsto p(\alpha)$$

This is called the 'evaluation' at  $\alpha$ . Clearly, it is a linear functional since

$$L_\alpha(c \cdot p(x) + q(x)) = c \cdot p(\alpha) + q(\alpha) = c \cdot L_\alpha(p(x)) + L_\alpha(q(x))$$

**Example.** Let  $C([a, b])$  be the space of continuous real-valued functions on  $[a, b]$ . Define

$$L : C([a, b]) \rightarrow \mathbb{R} : f \mapsto \int_a^b f(t) dt$$

$L$  is clearly a linear functional by the linearity of the integral.

**Definition 1.2.** The collection of all linear functionals on  $V$  is called the *dual space* of  $V$  and is denoted as  $V^*$ .

**Theorem 1.3.** Let  $V$  be a finite-dimensional vector space over  $F$ , and let  $\beta = \{\alpha_1, \dots, \alpha_n\}$  be a basis for  $V$ . Then, there exists a unique *dual basis*  $\beta^* = \{f_1, \dots, f_n\}$  for  $V^*$  such that  $f_i(\alpha_j) = \delta_{ij}$ . Furthermore, for each linear functional  $f \in V^*$ ,

$$f = \sum_{i=1}^n f(\alpha_i) f_i$$

and for each vector  $\alpha \in V$ ,

$$\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$$

*Proof.* By the property of basis, for any  $\alpha \in V$ , there exist unique scalars  $a_1, \dots, a_n \in F$  such that

$$\alpha = a_1 \alpha_1 + \dots + a_n \alpha_n$$

For each  $1 \leq i \leq n$ , define  $f_i : V \rightarrow F$  by  $f_i(\alpha) = a_i$  (or  $f_i(\alpha) = ([\alpha]_\beta)_i$ ).

Then  $f_i(\alpha_j) = \delta_{ij}$ . Since  $\dim V = \dim V^*$ , it is enough to show that the set  $\beta^* = \{f_1, \dots, f_n\}$  is linearly independent. Suppose that for some  $c_i \in F$ ,  $\sum_{i=1}^n c_i f_i = 0$ . Then for each  $1 \leq j \leq n$ ,

$$\left( \sum_{i=1}^n c_i f_i \right) (\alpha_j) = \sum_{i=1}^n c_i f_i(\alpha_j) = c_j = 0$$

Thus,  $\beta^*$  is linearly independent and forms a basis for  $V^*$ .

The second assertion can be proved as follows: if  $\alpha \in V$  is represented as  $\alpha = \sum_{i=1}^n x_i \alpha_i$  where  $x_i \in F$ , then for each  $1 \leq j \leq n$ :

$$f_j(\alpha) = f_j \left( \sum_{i=1}^n x_i \alpha_i \right) = \sum_{i=1}^n x_i f_j(\alpha_i) = x_j$$

Substituting  $x_j$  back into the representation of  $\alpha$ , we obtain  $\alpha = \sum_{i=1}^n f_i(\alpha) \alpha_i$ . □

## Application: Lagrange Interpolation

Let  $P_n(F) = \{p \in F[x] : \deg p \leq n\}$ , and let  $t_0, t_1, \dots, t_n \in F$  be distinct elements. For each  $0 \leq i \leq n$ , define the linear functional  $L_i : P_n(F) \rightarrow F$  by  $L_i(p) = p(t_i)$ . These  $\{L_0, L_1, \dots, L_n\}$  are linearly independent. Suppose that

$$c_0 L_0 + c_1 L_1 + \dots + c_n L_n = 0 \quad \dots (*)$$

where  $c_i \in F$ . By substituting the standard basis  $\{1, x, x^2, \dots, x^n\}$  into  $(*)$ , we obtain a system of equations:

$$\begin{cases} c_0 + c_1 + \dots + c_n = 0 \\ c_0 t_0 + c_1 t_1 + \dots + c_n t_n = 0 \\ \vdots \\ c_0 t_0^n + c_1 t_1^n + \dots + c_n t_n^n = 0 \end{cases} \implies \begin{bmatrix} 1 & 1 & \dots & 1 \\ t_0 & t_1 & \dots & t_n \\ \vdots & \vdots & \ddots & \vdots \\ t_0^n & t_1^n & \dots & t_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The matrix above is the **Vandermonde Matrix**, which is invertible since  $t_i$  are distinct. Thus  $c_0 = c_1 = \dots = c_n = 0$ . More precisely, define a linear map

$$T : P_n(F) \rightarrow F^{n+1} : p(x) \mapsto (p(t_0), \dots, p(t_n))$$

For  $\beta = \{1, x, \dots, x^n\}$ ,  $[T]_\beta$  is the **Vandermonde Matrix**.

If we choose a different basis  $\beta' = \{1, (x - t_0), (x - t_0)(x - t_1), \dots, \prod_{j=0}^{n-1} (x - t_j)\}$ , then  $[T]_{\beta'}$  becomes:

$$\det[T]_{\beta'} = \det \begin{bmatrix} 1 & 0 & \dots & 0 \\ 1 & t_1 - t_0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_n - t_0 & \dots & \prod_{j=0}^{n-1} (t_n - t_j) \end{bmatrix} = \prod_{1 \leq i < j \leq n} (t_j - t_i) \neq 0$$

Since the determinant is invariant with respects to the similar matrix, it is proved.

**Theorem 1.4.**  $\{L_0, \dots, L_n\}$  is a dual basis for  $P_n(F)^*$  with respect to the basis  $\{P_0(x), \dots, P_n(x)\}$  for  $P_n(F)$ , where

$$P_i(x) = \prod_{j \neq i} \frac{x - t_j}{t_i - t_j}$$

Since  $L_j(P_i) = P_i(t_j) = \delta_{ij}$ , for any  $p(x) \in P_n(F)$ , we have:

$$p(x) = \sum_{i=0}^n L_i(p) P_i(x) = \sum_{i=0}^n p(t_i) P_i(x)$$

Furthermore, by the linearity of the integral,  $\int_a^b p(x) dx = \sum_{i=0}^n p(t_i) \int_a^b P_i(x) dx$ .

## 2 Double Dual

*Question: Is every basis for  $V^*$  a dual of some basis for  $V$ ?*

**Lemma 2.1.** Let  $V$  be a finite dimensional vector space over  $F$ . For each  $\alpha \in V$ , define

$$\widehat{\alpha} : V^* \rightarrow F : f \mapsto f(\alpha)$$

Then, it is linear functional on  $V^*$ . Moreover, if  $\widehat{\alpha}(f) = 0$  for any  $f \in V^*$ , then  $\alpha = 0$ .

**Note:** the last assertion can be said:  $(V^*)^0 = \{0\}$ .

*Proof.* Let  $f, g \in V^*$  and  $c \in F$  be given. Then,

$$\widehat{\alpha}(cf + g) = (cf + g)(\alpha) = cf(\alpha) + g(\alpha) = c\widehat{\alpha}(f) + \widehat{\alpha}(g)$$

Thus proved. To show the second assertion, let  $\alpha \neq 0$ . Then, there exists a basis  $\beta$  for  $V$  containing  $\alpha$ . Then, there is a dual basis  $\beta^*$  for  $V^*$ . Take  $f_\alpha \in \beta^*$  corresponding to  $\alpha$ . Then,  $f_\alpha(\alpha) = 1 \neq 0$ . Thus proved.  $\square$

**Theorem 2.2.** Let  $V$  be a finite dimensional vector space over  $F$ . Then, the map

$$V \rightarrow V^{**} : \alpha \mapsto \widehat{\alpha}$$

is an Isomorphism.

*Proof.* Let  $\alpha, \beta \in V$  and  $c \in F$ , and for any  $f \in V^*$ ,

$$\widehat{(c\alpha + \beta)}(f) = f(c\alpha + \beta) = cf(\alpha) + f(\beta) = c\widehat{\alpha}(f) + \widehat{\beta}(f) = (c\widehat{\alpha} + \widehat{\beta})(f) = c\widehat{\alpha}(f) + \widehat{\beta}(f)$$

Moreover, by the Lemma, the kernel of the map is trivial. Thus injective, and surjective because:

$$\dim V^{**} = \dim V^* = \dim V$$

From the Theorem above, we obtain that:

If  $\widehat{\alpha} \in V^{**}$ , then there exists a unique  $\alpha \in V$  such that for all  $f \in V^*$ ,  $\widehat{\alpha}(f) = f(\alpha)$ .

We can restate this result as:

**Corollary 2.3.** Let  $V$  be a finite dimensional vector space over  $F$ . Every basis for  $V^*$  is the dual of some basis for  $V$ .

*Proof.* Let  $\{f_1, \dots, f_n\}$  be a basis for  $V^*$ . Then, there is a dual basis  $\{\widehat{\alpha}_1, \dots, \widehat{\alpha}_n\}$  for  $V^{**}$  such that  $\widehat{\alpha}_i(f_j) = \delta_{ij}$ . By the above corollary, for each  $\widehat{\alpha}_i \in V^{**}$ , there exists a unique  $\alpha_i \in V$  such that  $\widehat{\alpha}_i(f) = f(\alpha_i)$  for all  $f \in V^*$ . Now,  $\widehat{\alpha}_i(f_j) = f_j(\alpha_i) = \delta_{ji}$ , the claim is proved.  $\square$

**Theorem 2.4.** Let  $V$  be a finite-dimensional vector space, and let  $S \subset V$  be a subset. Then,

$$S^{00} = \widehat{\langle S \rangle}$$

where  $\widehat{\langle S \rangle}$  is natural isomorphic copy of  $\langle S \rangle$  in  $V^{**}$ .

### 3 Transpose

**Theorem 3.1.** Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $T : V \rightarrow W$  be a linear map. Then,

$$T^t : W^* \rightarrow V^* : g \mapsto g \circ T$$

is a linear map.

Furthermore, if  $V$  and  $W$  have finite dimensional with ordered basis  $\beta$  and  $\gamma$  for  $V$  and  $W$ , respectively, then

$$([T]_{\beta}^{\gamma})^t = [T^t]_{\gamma^*}^{\beta^*}$$

*Proof.* Since the composition of two linear maps is linear, the map  $T^t$  is well-defined. Given  $f, g \in W^*$  and  $c \in F$ ,

$$T^t(cf + g) = (cf + g) \circ T = cf \circ T + g \circ T = cT^t(f) + T^t(g)$$

Thus, the first assertion is proved.

Now, suppose that  $V$  and  $W$  have finite dimensional. Let  $\beta = \{x_1, \dots, x_n\}$  and  $\gamma = \{y_1, \dots, y_m\}$ .

Then, there are dual basis for  $V^*$ ,  $W^*$

$$\beta^* = \{f_1, \dots, f_n\}, \quad \gamma^* = \{g_1, \dots, g_m\}$$

respectively. Observe that: for each  $1 \leq j \leq m$ ,

$$T^t(g_j) = g_j \circ T = \sum_{s=1}^n (g_j \circ T)(x_s) f_s$$

Meanwhile, denote  $A = [T]_{\beta}^{\gamma}$ . Then,

$$(g_j \circ T)(x_i) = g_j(T(x_i)) = g_j\left(\sum_{k=1}^m A_{ki} y_k\right) = \sum_{k=1}^m A_{ki} g_j(y_k) = \sum_{k=1}^m A_{ki} \delta_{jk} = A_{ji}$$

Therefore,

$$[T^t]_{\gamma^*}^{\beta^*} = [[T^t(g_1)]_{\beta^*} \quad \dots \quad [T^t(g_m)]_{\beta^*}] = A^t = ([T]_{\beta}^{\gamma})^t$$

□

**Theorem 3.2.** Let  $F$  be a field, and let  $T : F^n \rightarrow F^m$  be a map. A map  $T$  is linear map if and only if:

There exist  $T_1, \dots, T_m \in (F^n)^*$  such that for all  $x \in F^n$ ,  $T(x) = (T_1(x), \dots, T_m(x))$

*Proof.* Let  $\{g_1, \dots, g_m\}$  be a dual basis for  $F^m$  with respects to standard basis for  $F^m$ . For each  $1 \leq i \leq m$ , define

$$T_i = T^t(g_i)$$

Let  $x \in V$  be given. Then, for each  $1 \leq i \leq m$ ,

$$T_i(x) = T^t(g_i)(x) = (g_i \circ T)(x) = g_i(T(x)) = ([T(x)]_{\gamma})_i$$

Thus proved. The converse is trivial.

□

## 4 Hyperspace

**Lemma 4.1.** Let  $f, g : V \rightarrow F$  be linear functionals on  $V$ .

$$g = c \cdot f \text{ for some } c \in F \text{ if and only if } \ker f \leq \ker g.$$

*Proof.* If  $f = 0$ , then it is trivial. Assume  $f \neq 0$ . Then  $\ker f$  is a hyperspace of  $V$ .

Let  $\alpha \in V$  such that  $f(\alpha) \neq 0$ . Put  $c = g(\alpha)/f(\alpha)$ .

Then,  $g - cf$  is a linear functional such that  $\ker f \leq \ker(g - cf)$  and  $\alpha \in \ker(g - cf)$ . Thus  $\ker(g - cf) = V$ , or  $g - cf = 0$ .  $\square$

**Theorem 4.2.** Let  $g, f_1, \dots, f_r$  be linear functionals on  $V$ . Then,

$$g \text{ is a linear combination of } \{f_1, \dots, f_r\} \text{ if and only if } \bigcap_{i=1}^r \ker f_i \leq \ker g.$$

*Proof.* If  $g = \sum_{i=1}^r c_i f_i$  for some  $c_i \in F$ , then for any  $\alpha \in \bigcap_{i=1}^r \ker f_i$ ,  $g(\alpha) = \sum_{i=1}^r c_i f_i(\alpha) = 0$ .

Conversely, we shall use the mathematical induction for  $r \in \mathbb{N}$ .

By the previous Lemma, if  $r = 1$ , it is proved. Suppose that it is held for  $r = k - 1$ .

Let  $f_1, \dots, f_k$  be linear functionals on  $V$  such that  $\bigcap_{i=1}^k \ker f_i \leq \ker g$ .

Let  $g, f_1, \dots, f_{k-1}$  be restricted by  $\ker f_k$ . That is,

$$f_i|_{\ker f_k} : \ker f_k \rightarrow F : \alpha \mapsto f_i(\alpha) \quad (i = 0, \dots, k-1, \text{ where } f_0 = g)$$

And, let  $\alpha \in \ker f_k$  such that for all  $1 \leq i \leq k-1$ ,  $f_i(\alpha) = 0$ .

That is,  $\alpha \in \bigcap_{i=1}^k \ker f_i$  and therefore  $g|_{\ker f_k}(\alpha) = 0$  by the hypothesis. By the assumption of the induction,

$$g|_{\ker f_k} = \sum_{i=1}^{k-1} c_i f_i|_{\ker f_k} \text{ for some } c_i \in F.$$

Put  $h = g - \sum_{i=1}^{k-1} c_i f_i$ . Now, for any  $\alpha \in \ker f_k$ ,  $h(\alpha) = 0$ , or  $\ker f_k \leq \ker h$ .

The previous Lemma gives that  $h = c_k f_k$  for some  $c_k \in F$ . Now,  $g = \sum_{i=1}^k c_i f_i$ .  $\square$

**Theorem 4.3.** Let  $V$  be a vector space over  $F$ .

For any non-zero linear functional  $f \in V^*$ ,  $\ker f$  is a hyperspace of  $V$ .

Conversely, every hyperspace in  $V$  is the kernel for some linear functional on  $V$ .

*Proof.* Let  $f \in V^*$  be a non-zero linear functional. Since  $f$  is non-zero, choose  $\alpha \in V \setminus \ker f$  (or  $f(\alpha) \neq 0$ ).

We shall show that  $V = \ker f + \langle \alpha \rangle$ . Let  $\beta \in V$  be given. Put  $c = f(\beta)/f(\alpha)$ . (It can be defined because  $f(\alpha) \neq 0$ ).

Then,  $f(\beta - c\alpha) = f(\beta) - cf(\alpha) = 0$ , thus  $\beta - c\alpha \in \ker f$ , or  $\beta \in \ker f + \langle \alpha \rangle$ .

Conversely, let  $N \leq V$  be a hyperspace. Let  $\alpha \in V \setminus N$ . Since  $N$  is a hyperspace,  $N + \langle \alpha \rangle = V$ . Now, for each  $\beta \in V$ ,

$$\beta = \gamma + c\alpha \text{ for some } \gamma \in N, c \in F.$$

This  $\gamma$  and  $c$  are uniquely determined because  $N \cap \langle \alpha \rangle = \{0\}$ . Define  $g : V \rightarrow F$  as  $g(\beta) = c$ . Then,  $\ker g = N$ .  $\square$

## 5 Annihilators

**Definition 5.1.** Let  $V$  be a vector space over  $F$ , and let  $S \subseteq V$  be a subset. The annihilator of  $S$  is the set

$$S^0 \stackrel{\text{def}}{=} \{f \in V^* \mid f(\alpha) = 0 \text{ for all } \alpha \in S\}.$$

**Note:**  $f \in S^0 \iff S \subseteq \ker f$ .

**Theorem 5.2.** Let  $V$  be a finite-dimensional vector space, and let  $W \leq V$  be a subspace. Then

$$\dim W + \dim W^0 = \dim V.$$

*Proof.* Clearly,  $W^0 \leq V^*$ . Let  $\{\alpha_1, \dots, \alpha_k\}$  be a basis for  $W$ .

Choose  $\alpha_{k+1}, \dots, \alpha_n \in V$  such that  $\{\alpha_1, \dots, \alpha_n\}$  is a basis for  $V$ .

Let  $\{f_1, \dots, f_n\}$  be the dual basis for  $V^*$  corresponding to  $\{\alpha_1, \dots, \alpha_n\}$ .

For  $k+1 \leq i \leq n$ , we have  $f_i(\alpha_j) = 0$  for all  $1 \leq j \leq k$ . Thus,  $f_i(\alpha) = 0$  for all  $\alpha \in W$ , which implies  $\{f_{k+1}, \dots, f_n\} \subseteq W^0$ .

Clearly, this set is linearly independent. To show that it spans  $W^0$ , let  $f \in W^0 \leq V^*$  be given. Then

$$f = \sum_{i=1}^n f(\alpha_i) f_i = \sum_{i=k+1}^n f(\alpha_i) f_i \in \text{span}\{f_{k+1}, \dots, f_n\}$$

since  $f(\alpha_i) = 0$  for  $1 \leq i \leq k$ . Thus,  $\dim W^0 = n - k = \dim V - \dim W$ . □

**Corollary 5.3.** If  $W \leq V$  with  $\dim W = k < n = \dim V$ , then

$$W = \ker f_{k+1} \cap \dots \cap \ker f_n$$

where  $f_i$  are the functionals described in the theorem above.

**Corollary 5.4.** For subspaces  $W_1, W_2 \leq V$  with  $\dim V < \infty$ ,

$$W_1 = W_2 \iff W_1^0 = W_2^0.$$

## 6 Properties of Annihilators

**Proposition 6.1.** Let  $V$  be a vector space over  $F$ , and let  $W_1, W_2 \leq V$  be subspaces. Then:

1.  $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$
2.  $(W_1 \cap W_2)^0 = W_1^0 + W_2^0$

*Proof.* 1. is trivial. To show 2., let  $f \in (W_1 \cap W_2)^0$ . Define a map

$$f_1 : W_1 + W_2 \rightarrow F, \quad \alpha_1 + \alpha_2 \mapsto f(\alpha_2)$$

It is well-defined because  $\alpha_1 + \alpha_2 = \alpha'_1 + \alpha'_2$  implies  $\alpha_2 - \alpha'_2 = \alpha'_1 - \alpha_1 \in W_1 \cap W_2$ , and since  $f \in (W_1 \cap W_2)^0$ , we have

$$f(\alpha_2 - \alpha'_2) = f(\alpha_2) - f(\alpha'_2) = 0$$

which gives  $f(\alpha_2) = f(\alpha'_2)$ .

Now, for a basis  $\beta$  of  $W_1 + W_2$  and an extension  $\beta \cup \beta'$  for  $V$ , define an extension  $\overline{f}_1 : V \rightarrow F$  such that:

$$\overline{f}_1(\alpha) = \begin{cases} f_1(\alpha) & \text{if } \alpha \in \langle \beta \rangle \\ 0 & \text{if } \alpha \in \langle \beta' \rangle \end{cases}$$

Put  $f_2 = f - \overline{f}_1$ . Then  $f = \overline{f}_1 + f_2$ ,  $\overline{f}_1 \in W_1^0$ , and  $f_2 \in W_2^0$ , thus  $(W_1 \cap W_2)^0 \subseteq W_1^0 + W_2^0$ . The converse is trivial.  $\square$

**Theorem 6.2.** Let  $W \leq V$  and let  $\{g_1, \dots, g_r\}$  be a spanning set for  $W^0$ . Then

$$W = \bigcap_{i=1}^r \ker g_i$$

*Proof.* The inclusion  $W \subseteq \bigcap_{i=1}^r \ker g_i$  is trivial, because by the definition of the annihilator,  $W \subseteq \ker g$  for all  $g \in W^0$ .

Conversely, let  $\{f_1, \dots, f_m\} \subseteq V^*$  be a set of linear functionals such that  $W = \bigcap_{i=1}^m \ker f_i$ . Then for each  $1 \leq i \leq m$ ,

$$W \subseteq \ker f_i \iff f_i \in W^0.$$

Since  $\{g_1, \dots, g_r\}$  is a spanning set for  $W^0$ , each  $f_i$  is a linear combination of  $\{g_1, \dots, g_r\}$ .

By the previously proved thm 4.2, this implies:

$$\bigcap_{j=1}^r \ker g_j \subseteq \ker f_i \quad \text{for each } i = 1, \dots, m.$$

Taking the intersection over all  $i$ , we obtain:

$$\bigcap_{j=1}^r \ker g_j \subseteq \bigcap_{i=1}^m \ker f_i = W.$$

Thus, the equality is proved.  $\square$

## 7 Transpose and Annihilators

**Theorem 7.1.** Let  $V$  and  $W$  be vector spaces over  $F$ , and let  $T: V \rightarrow W$  be a linear map. Then

$$\ker T^t = (\operatorname{Im} T)^0.$$

In particular, if  $V$  and  $W$  are finite-dimensional, then:

1.  $\operatorname{rank} T^t = \operatorname{rank} T$
2.  $\operatorname{Im} T^t = (\ker T)^0$

*Proof.* The first assertion is proved as follows:

$$g \in \ker T^t \iff T^t(g) = 0 \iff g(T(\alpha)) = 0 \text{ for all } \alpha \in V \iff g(\gamma) = 0 \text{ for all } \gamma \in \operatorname{Im} T \iff g \in (\operatorname{Im} T)^0.$$

Thus,  $\ker T^t = (\operatorname{Im} T)^0$ . Now, suppose that  $V$  and  $W$  have finite-dimensional.

Put  $n = \dim V$  and  $m = \dim W$  and  $r = \operatorname{rank} T$ . Since  $\dim \operatorname{Im} T + \dim (\operatorname{Im} T)^0 = \dim W$ ,  $\dim (\operatorname{Im} T)^0 = m - r$ .

Since the first result of this theorem with the Dimension Theorem,

$$m - r = \dim \ker T^t = \dim W^* - \dim \operatorname{Im} T^t = m - \operatorname{rank} T^t$$

Therefore,  $\operatorname{rank} T^t = m$ . To show the last assertion, let  $f \in \operatorname{Im} T^t$ . That is,  $f = T^t(g)$  for some  $g \in W^*$ . Now,

$$f(\alpha) = T^t(g)(\alpha) = (g \circ T)(\alpha) = g(0) = 0 \text{ for all } \alpha \in \ker T$$

Thus  $f \in (\ker T)^0$ , or  $\operatorname{Im} T^t \subseteq (\ker T)^0$ . Finally,

$$\dim (\ker T)^0 = \dim V^* - \dim \ker T = \operatorname{rank} T = \operatorname{rank} T^t$$

Proved. □

## 8 System of Linear Equations in the View of Linear functionals

Consider a system of linear equations:

$$\begin{cases} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n = 0 \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n = 0 \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n = 0 \end{cases}$$

For each  $1 \leq i \leq m$ , define a linear functional  $f_i : F^n \rightarrow F$  as:

$$f_i(x_1, \dots, x_n) = \sum_{j=1}^n A_{ij}x_j$$

Then, solving the system of equations is equivalent to finding the subspace  $W \subseteq F^n$ :

$$W = \ker f_1 \cap \ker f_2 \cap \cdots \cap \ker f_m = \bigcap_{i=1}^m \ker f_i$$

Using the natural identification,

$$W = \bigcap_{i=1}^m \ker f_i = \left( \sum_{i=1}^m (\ker f_i)^0 \right)^0 = \left( \sum_{i=1}^m \langle f_i \rangle \right)^0 = \{f_1, \dots, f_m\}^0$$

That is, the solution set is exactly the Annihilator of the row space.

Let  $g_1, \dots, g_r$  be a basis for  $\langle f_1, \dots, f_m \rangle$ . Then clearly  $r \leq m$ .

And, choose  $g_{r+1}, \dots, g_n \in V^*$  such that  $g_1, \dots, g_n$  be a basis for  $V^*$ . Now, there exists a basis  $\{\alpha_1, \dots, \alpha_n\}$  for  $V$  s.t

$$g_i(\alpha_j) = \delta_{ij}$$

Finally,  $\{\alpha_{r+1}, \dots, \alpha_n\}$  be a basis for  $W$ .

Conversely, suppose that  $W \leq F^n$  is a subspace. How can we find the linear system having  $W$  as a solution set?

## 9 Inner Product Space and Linear functional

Recall that:

**Theorem 9.1.** Let  $V$  be a finite dimensional inner product vector space over  $F$ .  
For any linear functional  $g: V \rightarrow F$ , there exists a unique vector  $y \in V$  such that

$$g(x) = \langle x, y \rangle, \quad \forall x \in V$$